

Abhishek Tagalpallewar

Junior Undergraduate
Mechanical Engineering, Minors in Computer Science and Engineering
Indian Institute of Technology Gandhinagar

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ACADEMIC DETAILS

Degree	Institution	Year	CPI/%
B.Tech	Indian Institute of Technology Gandhinagar	2024-2028	8.70
Class XII	Late P. B. Jog Junior College (MSBSHSE)	2024	89.5
Class X	Mahesh Vidyalaya English Medium School	2022	95.8

PROJECTS

Neural Network Framework for Learning Constitutive Laws in 2D Granular Flow [Jan'26-Apr'26]
(Advisor - Prof. Harkirat Singh, IIT Gandhinagar) | Project Link

- Engineered a modular **PyTorch** deep learning framework, evaluating **3** critical internal stress components (σ_{xx} , σ_{yy} , σ_{xy}) across **25** hyperparameter configurations to map nonlinear physics paths
- Reduced overall cross-dataset prediction error rates from **48%** down to **3%** by implementing reference normalization models, confirming robust generalization on **2** unseen datasets
- Automated **2D spatial grid mapping** for large-scale internal **2D stress tensor** variations, streamlining the data preprocessing and training pipeline

SmartSpend AI – Full Stack Personal Finance Tracker with AI Budget Coach [May'26-Jun'26]
Project Link | Live Demo

- Built a responsive full-stack finance web application, executing automated tracking logic across **100+** personal entries to evaluate custom categorical budgetary allocation thresholds
- Integrated the conversational **Google Gemini API** with a robust **React**, **Node.js**, **Express**, and **MongoDB** backend architecture to deliver personalized budgeting advice across **5** financial categories
- Structured backend tracking across **3** distinct data schemas, engineering secure user authentication and interactive dashboards

Validation of Angular Momentum Conservation via Reynolds Transport Theorem [Jan'26-Apr'26]
(Advisor - Prof. Shubhangi Bansude, IIT Gandhinagar) | Project Link

- Validated physical milestones by fabricating a custom **3-nozzle** rotating water sprinkler mechanism prototype from scratch
- Modeled relative error variances of **31.1%** and **42.0%** across **2** operating speeds by formulating a fluid predictive framework

RELEVANT COURSEWORK

- Data Structures and Algorithms | Probability, Statistics and Data Visualization | Calculus of Single Variable and Linear Algebra

TECHNICAL SKILLS

- Programming Languages:** Python | C | C++
- Developer Tools:** VS Code | Git | GitHub
- Libraries:** NumPy | Pandas | Matplotlib | PyTorch
- Software & Simulation:** MATLAB | Cantera | Mathematica | Autodesk Inventor | MSC Adams (Basic) | ANSYS Fluent (Basic) | Creality Print

ACHIEVEMENTS

- Secured **Dean's List** (top 5%) and **Rank 1 (Branch Topper)** in Mechanical Engineering with **9.48 SPI** in Semester 3
- Earned **A+ (11/10)** in **Thermodynamics** for exceptional performance and accuracy across advanced curriculum evaluation structures
- Won Division **4** with the chess team at India Chess Championship 2025 (**250+** teams across **30** divisions)
- Claimed **Team Runner-up** at DA-IICT Chess Tournament and individual **Runner-up (Unrated Category)** at IIT Bombay Sports Fest

POSITIONS OF RESPONSIBILITY

- Coordinating Team Member | Institute Sports Events, IIT Gandhinagar** [Jul'24-Apr'26]
 - Managed basic operational logistics, scheduling, and tracker loops for **3+** major sports festivals
 - Collaborated closely with a **15-member** student council to streamline tournament timelines
- Junior Team Member | Amalthea & Blithchron Festivals, IIT Gandhinagar** [Aug'24-Feb'25]
 - Coordinated core event operations during Amalthea'25 by assisting with hospitality logistics
 - Tracked company outreach and record matrices to maintain partner communications for Blithchron'25

EXTRA-CURRICULAR ACTIVITIES

- Represented IIT Gandhinagar twice at the **Inter IIT Sports Meet**, competing in Chess matches against **20+** IITs
- Participated in chess tournaments and inter-university matches hosted at Ahmedabad University and IIT Jodhpur